



CODECHEF

The CodeChef.com Challenge, your monthly dose of coding puzzles from India's biggest coding contest, is now in print!

Hello and welcome to the CodeChef Puzzle Challenge sponsored by *LINUX For You* and Direct! Let's see how many of you can solve this month's puzzle and go on to become one of the three lucky winners to bag some awesome prizes, both in cash and merchandise!

The July edition's puzzle

If you recall, the May 2011 puzzle was about a rich merchant who had 32 gold coins. This month, he is in urgent need of our help. He has placed his gold coins in a locker, only to realise that he has forgotten the number combination that opens the locker! He however finds five clues that can help him remember the code:

1. The 2nd number plus the 3rd number equal 10.
2. The 1st number is one less than twice the 2nd number.
3. The 4th number is one more than the 2nd number.
4. The 5th number plus the 3rd number equal 14.
5. The sum of all 5 numbers is 30.

What were the 5 numbers? Arrange them in the order wherein they would work as the locker's password.

The solution

The number combination to unlock the rich merchant's safe is 7 4 6 5 8.

How did we do it?

Let the five numbers be A, B, C, D and E.
Given that the sum of the 5 numbers is 30, the first equation will be as follows:
 $A + B + C + D + E = 30$ (Equation 1)
Following the rest of the conditions, we arrive at the following equations:

- $B + C = 10$ (Equation 2)
- $A = 2B - 1$ (Equation 3)
- $D = B + 1$ (Equation 4)
- $E + C = 14$ (Equation 5)

Solving Equations (2) and (5) we get $B - E = -4$ which can also be written as:

$$E = B + 4 \text{ (Equation 6)}$$

Re-writing Equation 1 in terms of B, we get:

$$2B - 1 + 10 + B + 1 + B + 4 = 30 \text{ (since } A = 2B - 1, B + C = 10, D = B + 1 \text{ and } E = B + 4).$$

$$\text{Hence, } B = 4.$$

By substituting B with its real value in the rest of the equations, we get the remaining values.

And the winners this month are:

- Sushant Shambharkar
- Mitul Maniar
- Arun Kumar

Your CodeChef Puzzle Of The Month

An artist has created a huge pile of different paintings and is exhibiting his works of art in a gallery. Periodically, he picks up one of the paintings and hangs it on the wall, and adjusts it.

- At 11:00 am, the painting titled 'Number 30' was hung.
- At 4:00 pm, the painting titled 'Number 240' was hung.
- At 7:30 pm, the painting titled 'Number 315' was hung.

What will be the number of the painting that the artist will hang on the wall at 9:20 pm?

Looking for exciting prizes?

Solve this month's CodeChef Challenge puzzle and send in your answer by the August 15, 2011, to codechef@efyindia.com. You can also post your answer to the puzzle online at <http://www.linuxforu.com/codechef/>. Three lucky winners will get exciting prizes, both in cash and merchandise! **END** 

About CodeChef: CodeChef.com is India's first, non-commercial, online programming competition, featuring monthly contests in more than 35 different programming languages. Log on to CodeChef.com for the August Challenge that runs from August 1 to 11, to win cash prizes of up to Rs 20,000. But that's not all! Stay tuned to CodeChef and participate in the multiple programming contests that take place on the website throughout the month!